Sangyoung Son, Ph.D.

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Education

University of Southern California, Los Angeles, CA, USA — Aug. 2012
Ph.D. in Civil Engineering (Advisor: Patrick J. Lynett)
– Dissertation: *Wave-Induced Hydrodynamic Complexity and Transport in the Nearshore*

Texas A&M University, College Station, TX, USA — Dec. 2011
M.E. in Civil Engineering, non-thesis (Advisor: Patrick J. Lynett, relocated to USC)
– Research topic: *Coupled Numerical Model for Lifeline Tsunami Evolution*

Seoul National University, Seoul, Korea — Feb. 2002
M.S. in Civil, Urban and Geosystem Engineering (Advisor: Kyung-Duck Suh)
– Thesis: *Calculation of Irregular Wave Reflection from Perforated-Wall Caisson Breakwaters Using a Regular Wave Model*

Seoul National University, Seoul, Korea — Feb. 2000
B.S. in Civil, Urban and Geosystem Engineering

Employment

Korea University, Seoul, Korea — Sep. 2015 – Present
Professor — 2023 – Present
Associate Professor — 2018 – 2023
Assistant Professor — 2015 – 2018

Dohwa Engineering Co., Ltd., Seoul, Korea — Mar. 2026 – Present
Independent (Non-executive) Director

University of Southern California, Los Angeles, CA, USA — Jan. 2019 – Aug. 2019
Visiting Faculty

University of Ulsan, Ulsan, Korea — Sep. 2013 – Aug. 2015
Assistant Professor

University of Southern California, Los Angeles, CA, USA — Sep. 2012 – Aug. 2013
Postdoctoral Scholar – Research Associate

University of Southern California, Los Angeles, CA, USA — Sep. 2011 – Aug. 2012
Graduate Research Assistant

Oregon State University, Corvallis, OR, USA — May 2009 – Jul. 2009
Visiting Graduate Researcher, O.H. Hinsdale Wave Research Laboratory
– Laboratory experiments on solitary wave evolution over a 3D shallow shelf (funded by NSF)

Texas A&M University, College Station, TX, USA — Aug. 2008 – Aug. 2011
Graduate Research Assistant

Hyundai Engineering & Construction Co., Ltd., Korea — Mar. 2002 – Jul. 2008
Assistant Senior Engineer

Research Interests

My group studies the physics of wave- and current-driven coastal flows and develops fast, high-fidelity simulation and AI tools for predicting and preparing for coastal hazards such as tsunamis, storm surges and compound floods. Bracketed numbers refer to Journal Publications (J) and Manuscripts under Review (R).

GPU-accelerated, interactive and immersive coastal modeling
– Real-time GPU and virtual-reality models for nearshore waves, tsunamis, scalar transport and flooding, built on robust numerics for dispersive shallow water flows: well-balanced wet/dry reconstruction, contact-discontinuity capturing and adaptive time integration [J43, J42, J38, J35, J30].

Tsunami generation, propagation and impacts
– Kinematic and multi-fault rupture, adjoint source inversion, nested multi-physics modeling, runup physics, and tsunami-induced currents, debris and scour in harbors, supported by post-event field surveys [J49, J40, J34, J32, J26, J23, J21, J18, J16, J8, J5, J4, J3, J2, R10].

Storm surge, compound flooding and climate change
– Controls of tropical cyclone motion, intensification and coastal geometry on surge extremes; projections of future surge, wave and tsunami hazard; and coupled atmosphere–hydrology–hydrodynamic modeling of tropical cyclone rainfall and compound flooding [J46, J45, J44, J39, J36, J25, J24, J17, R2, R3, R4, R7].

AI and satellite data for coastal hazards
– Machine-learning prediction of tropical cyclone tracks, intensity and wind/pressure fields, satellite-based wind field reconstruction, neural-operator and deep-learning surrogates, coupling of AI weather forecasts with hydrodynamic models, and optimal tsunami gauge deployment [J48, J47, J41, J33, J29, R1, R5, R6].

Wave hydrodynamics and nearshore processes
– Dispersive waves over sheared currents, stratified fluids and viscous mud; partially reflecting structures and wavemakers; rip currents, tidal vortices, turbulence, oil spill transport and morphological change [J37, J31, J28, J27, J22, J20, J19, J15, J14, J13, J12, J11, J10, J9, J7, J6, J1, R8, R9].

Graduate Supervision

Ph.D.: 3 graduated, 2 current M.S.: 8 graduated, 3 current Postdoctoral researchers: 2 former, 3 current

Graduated Ph.D.

Asrini Chrysanti (2025) – *Impact of Ocean and Land Surface Changes on Atmospheric Interactions and Extreme Storms*; now Lecturer, Bandung Institute of Technology, Indonesia

Xiaojuan Qian (2025) – *The Impact of Tropical Cyclone Motion on Storm Surge and Compound Flooding*; now Assistant Professor, Zhejiang Ocean University, China

Sooncheol Hwang (2023) – *Immersive and Interactive Multi-Physics Modeling of Coastal Hazards*; now Senior Researcher, Korea Institute of Ocean Science and Technology (KIOST)

Graduated M.S.

Mijin Kim (2026) – *Projection of Future Wave Climate Using High-Resolution GCMs*; now Engineer, Korea Ports & Harbors Association

Yangheon Song (2026) – *Modelling of Volcanic Tsunami Generation in the Surface-Piercing and Submerged Regimes*; now Researcher, Korea University

Vera Lucia Villanueva Roman (2025) – *Integrating AI-based Weather Forecasts with Hydrodynamic Modeling*; now Manager, Hyundai Engineering & Construction

Sehyuk Im (2025) – *A Study on Estimating Typhoon Wind Fields Using Satellite Imagery*

Jihyeon Yoon (2025) – *A Prediction Technique for Tropical Cyclone Pressure Fields Using Random Forest*; now Researcher, Oceanic

Junsu Noh (2023) – *Development of an Interactive Sediment Transport Model Based on GPU Acceleration*; now Ph.D. student, Korea University

Yongsu Na (2019) – *Forecasting of Tropical Cyclone Track and Intensity Using Echo State Networks*; Ph.D., HKUST; now Postdoctoral Researcher, Okinawa Institute of Science and Technology

Chilwoo Lee (2018) – *Coupled Storm Surge–Tide–Riverine Flow Modelling*; now Engineer, Yooshin Engineering Co.

Former Postdoctoral Researchers

Byoungjoon Na (2019 – 2021) – Experimental study on air–sea interaction; now Assistant Professor, Kumoh National Institute of Technology

Jung-A Yang (2018 – 2019) – Modelling morphological changes in the coast using Delft3D; now Postdoctoral Researcher, Korea University

Current Members

Postdoctoral researchers / researchers: Jin-Young Kim (Ph.D., Texas A&M, since 2024), Myung Jin Koh (Ph.D., University of Hawaii, since 2025), Yangheon Song (M.S., Korea University, since 2026)

Ph.D. students: Junsu Noh (since 2024), Jewon Kang (integrated M.S.–Ph.D., since 2025)

M.S. students: Seungtaek Oh (since 2025), Eddy Kwoh Zhi Yan (since 2026, GKS scholar), Shubh Mittal (since 2026, Seoul Tech scholarship)

Teaching at Korea University

Graduate: ACE 635 Computational Hydraulics; ACE 564 Hydrodynamics; ACE 832 Tidal and Coastline Hydraulics

Undergraduate: ACEE 228 Fluid Mechanics and Laboratory; ACEE 338 Coastal and Harbor Engineering; ACEE 348 Coastal Disaster Modelling; EGRN 151 Computer Language and Laboratory

Research Grants (Principal Investigator)

Real-time immersive compound flood model using extended reality (XR) technology
National Research Foundation of Korea (NRF), Mid-Career Researcher Program 2024 – 2026

Mechanisms and prediction technology for marine hazardous weather (with Pukyong National Univ.)
Korea Meteorological Administration, Hazardous Weather R&D 2025 – Present

Storm surge vulnerability under sea-level rise using future ocean modeling and AI emulators
NRF, Brain Pool Program (international researcher invitation) 2024 – 2025

Optimization and effect analysis of flood response technology reflecting watershed characteristics
KEITI, Climate-Crisis Flood Defense Innovation Program 2023 – 2025

Immersive wave and flow model based on a 3D virtual reality architecture
NRF, Mid-Career Researcher Program 2019 – 2024

Interactive flood routing model based on 3D virtual reality
Korea Water Resources Corporation (K-water) 2021 – 2022

Field combined PIV study of physical air–sea interaction at the sea surface
NRF, Brain Pool Program (international young researcher invitation) 2019 – 2021

Faster-than-real-time, GPU-accelerated interactive integrated inundation model
NRF, Basic Research Program 2017 – 2020

Tsunami prediction technology considering fault rupture history
Korea Meteorological Industry Promotion Agency, Weather Seed Technology Program 2018 – 2020

Practical coastal erosion management and response technology
Korea Institute of Marine Science & Technology Promotion (KIMST) 2018 – 2019

Storm surge assessment for design-basis and beyond-design-basis sea levels at nuclear power plants
Korea Institute of Nuclear Safety (KINS), Nuclear Safety R&D 2018

Scenario-based inundation mapping linking inland, riverine and coastal flooding from compound causes
Ministry of Public Safety and Security, Natural Disaster Prediction and Mitigation R&D 2016 – 2018

Multi-grid, multi-dimensional, high-efficiency wave transformation model using parallel computing and two-way coupling
NRF, Young Researcher Program 2014 – 2017

High-resolution pollutant transport prediction model for accidental spills in Ulsan coastal waters
Ulsan Green Environment Center 2015

Other projects: Development of a nonlinear tsunami inundation model (I), National Disaster Management Research Institute (2026); climate-resilient safe-port guidelines for ASEAN member states, Ministry of Oceans and Fisheries / Korea Ports and Harbours Association (2025 – Present); overtopping barrier technology for rapid flood response, Ministry of Climate, Energy and Environment (2026 – Present). Participating faculty in the BK21 PLUS (2016 – 2020) and BK21 FOUR (2020 – Present) programs, Ministry of Education.

Editorial Service for SCIE-indexed Journals

Section Editor, **Geoscience Letters** Dec. 2023 – Present

Associate Editor, **Journal of Hydro-environment Research** Mar. 2022 – Present

Topical Editor, **Applied Sciences** Jul. 2021 – Present

Associate Editor, **Coastal Engineering Journal** Jul. 2019 – Present

Guest Editor, **Journal of Marine Science and Engineering** 2019 – 2020
Special Issue "Advances in Nearshore Hydrodynamics Research" (5 papers)

International Conference Organization and Service

Session Convener May 2026
39th International Conference on Coastal Engineering (ICCE 2026), Galveston, TX, USA

Local Organizing Committee and Session Chair (Nearshore Currents and Sediment) Nov. 2025
International Conference on Asian and Pacific Coasts (APAC 2025), Busan, Korea

Local Organizing Committee, Seoul, Korea Jul. 2021
9th International Symposium on Environmental Hydraulics (ISEH)

Organizer, Korea University, Seoul, Korea Oct. 2019
International Workshop on Advanced Technology in Coastal Engineering

International Coastal Engineering Committee & TPC Member 2018 – 2019
International Ocean and Polar Engineering Conference (ISOPE)

Local Organizing Committee, Busan, Korea May 2018
International Coastal Symposium (ICS)

Local Organizing Committee, Incheon, Korea Aug. 2016
12th International Conference on Hydroinformatics (HIC)

Invited Talks and Expert Lectures

51st ASEAN MTWG Meeting, Phnom Penh, Cambodia Oct. 2026
Progress of Korea–ASEAN safe-port cooperation projects (Korean delegation)

Port Expert Workshop for Korea–ASEAN Cooperation (MOF/KPHA), Seoul, Korea Jul. 2026
Methods and Technological Advances in Assessing Coastal Vulnerability to Climate Change

UN ESCAP Capacity Building Workshop on Small Ports, Bangkok, Thailand Dec. 2025
Assessment Methods and Technological Trends for Evaluating Coastal Vulnerability to Climate Change

Workshop on Safe Ports in Response to Climate Change (MOF/ASEAN), Seoul, Korea Jul. 2025
Increasing Risk of Tropical Cyclones in Coastal Areas

48th ASEAN MTWG Meeting, Bandar Seri Begawan, Brunei May 2025
Progress of Korea–ASEAN safe-port cooperation projects (Korean delegation)

Universitas Hasanuddin, Indonesia (virtual guest lecture) Oct. 2024
Shallow Flow Modelling Based on Perturbation Theory: Past, Present, and Future

Int. Symposium on Hydro-Climate, Universiti Teknologi Malaysia Jul. 2024
Unveiling the Role of Cyclone Movement in Storm Surge Dynamics

International Research Visits and Workshops

GFZ Helmholtz Centre for Geosciences, Potsdam, Germany — May – Jun. 2026
Geo-INQUIRE workshop on probabilistic tsunami risk analysis (PTRA)

University of Tokyo, Japan (Prof. Takenori Shimozono) — Nov. 2025
Korea–Japan collaborative program on sustainable coastal monitoring technologies (KOFST/KWRA)

Disaster Prevention Research Institute, Kyoto University, Japan (Prof. Nobuhito Mori) — Oct. 2018
Collaboration on climate projection data for long-term shoreline change prediction

eCoast Marine Consulting and Research, Raglan, New Zealand (Dr. Jose C. Borrero) — Jan. 2017
Tsunami modelling for ports and harbours with COMCOT and MOST

Honors and Awards

Minister's Commendation — Mar. 2025
Minister of Land, Infrastructure and Transport, Korea

Academic Award — Jan. 2024
Korea Water Resources Association, Korea

Academic Award — Feb. 2022
Korean Society of Hazard Mitigation, Korea

President's Commendation — Dec. 2020
Korean Society of Civil Engineers, Korea

President's Commendation — May 2020
Korea Water Resources Association, Korea

Best Paper Award, KOSHAM Annual Conference — Feb. 2019
Korean Society of Hazard Mitigation, Korea

Minister's Commendation for Supporting Services of Hazard Mitigation — Jan. 2019
Ministry of the Interior and Safety, Korea

Best Paper Award, KSCE Annual Conference — Oct. 2018
Korean Society of Civil Engineers, Korea

Best Presentation Award, KWRA Annual Conference — May 2017
Korea Water Resources Association, Korea

Best Paper Award, KSCDP Annual Conference — Nov. 2015
Korean Society of Coastal Disaster Prevention, Korea

Best Paper Award, KSCOE Fall Conference — Oct. 2013
Korean Society of Coastal and Ocean Engineers, Korea

Chester P. Jelesnianski Scholarship — 2010 – 2011
Zachry Department of Civil Engineering, Texas A&M University

Joseph A. Orr Fellowship — 2008 – 2009
Zachry Department of Civil Engineering, Texas A&M University

Scholarship — 1997 – 1999
Seoul National University, Korea

Certificates

Professional Engineer (#79108PE), Oregon, USA since 2007

Information Processing Engineer, Korea since 1999

Patents

S. Son, Tsunami simulation method considering the time history of fault rupture
Korean Patent No. 10-2794042 (registered; filed Dec. 2021) Apr. 2025

S. Son, J. Yoon, Apparatus for estimating the pressure field of a typhoon
Korean Patent Application No. 10-2024-0198963 Dec. 2024

Books and Translations

Design of Movable Weirs and Storm Surge Barriers (Korean translation; lead translator, 2 translators)
Korea Waterway Transport Facilities Association, ISBN 979-11-88259-01-4 Apr. 2017

Professional Memberships

American Society of Civil Engineers (ASCE) | Coasts, Oceans, Ports and Rivers Institute (COPRI) | American Geophysical Union (AGU) | European Geosciences Union (EGU) | International Association for Hydro-Environment Engineering and Research (IAHR) | Coastal Education & Research Foundation (CERF) | International Society of Offshore and Polar Engineers (ISOPE) | Korean Society of Civil Engineers (KSCE) | Korea Water Resources Association (KWRA) | Korean Society of Coastal and Ocean Engineers (KSCOE) | Korean Society of Hazard Mitigation (KOSHAM) | Korean Wetland Society (KWS) | Korean Society of Ocean Engineers (KSOE)

Journal Publications

† denotes graduate/post-doctoral advisee.

J49. M.-J. Koh†, H. Park, **S. Son** (2026) Stochastic Tsunami-Driven Debris Hazard in a Built Environment: Community-Scale Assessment at Kahului Harbor, Maui, ***Coastal Engineering***, 212, 105131.

J48. J. Kim, D. Wee, Y. Jang, J. Choi, H. Hong, **S. Son** (2026) Strategic Offshore Deployment of Tsunami Gauges for Rapid Warning at Critical Coastal Sites: A Case Study for a Nuclear Power Plant in Korea, ***Applied Ocean Research***, 175, 105253.

J47. J. Kim†, **S. Son** (2026) A Boundary-Informed UNet++–ConvLSTM Emulator of a SWAN-based Delft3D-FM WAVE Hindcast for Nearshore Wave-Field Reconstruction, ***Applied Ocean Research***, 174, 105211.

J46. X. Qian†, **S. Son** (2026) Separating Typhoon and Non-typhoon Rainfall Reveals Structural Hydrological Regime Shifts in Coastal-to-Inland Basins, ***Journal of Hydrology***, 677 (Part C), 135941.

J45. H. O. Parmes, A. Alcantara, **S. Son**, K.-H. Ahn (2026) Rapid Intensification Restructures the Upper Tail of Tropical-Cyclone Storm Surge Extremes, ***Environmental Research Letters***, 21, 124010.

J44. A. Chrysanti†, **S. Son** (2026) Large-Scale Drivers and Ocean–Land Feedbacks Contributing to Extreme Precipitation During the January 2021 South Kalimantan Flood, Indonesia, ***Weather and Climate Extremes***, 52, 100894.

J43. S. Hwang†, P. Lynett, **S. Son** (2026) A Second-order Well-balanced Reconstruction for Shallow Flows with Wet/dry Fronts, ***Computers and Mathematics with Applications***, 208, 33–54.

J42. P. Lynett, B. Ebrahimi, **S. Son**, S. Hwang†, S. Bak (2026) An Interactive Nearshore Wave Simulator for Rapid Design Prototyping and Natural Hazard Education, ***ASCE Journal of Waterway, Port, Coastal, and Ocean Engineering***, 152(4), 04026007.

J41. J. Youn, S. Hwang, X. Qian, S. Im, **S. Son** (2026) A Random Forest-based Method for Reconstructing Pressure Fields in Tropical Cyclones from Wind Data, ***Atmospheric Research***, 330, 108588.

J40. D.-H. Kim, **S. Son**, T.-H. Jung (2025) Topocaustically Driven Similarity in Long-Distance Tsunami Runup along a Coast, ***KSCE Journal of Civil Engineering***, 29(9), 100194.

J39. A. Chrysanti†, **S. Son** (2025) Unraveling Atmosphere and Surface Boundary Interactions Behind Extreme Tropical Rainfall: A Case Study in Indonesia Using Fully Coupled Atmosphere–Hydrology Simulations, ***Journal of Advances in Modeling Earth Systems***, 17(4), e2024MS004730.

J38. S. Hwang†, P. Lynett, **S. Son** (2025) A GPU-accelerated Numerical Model for Nearshore Scalar Transport by Dispersive Shallow Water Flows, ***Computer Physics Communications***, 310, 109539.

J37. S. Hwang, B. Na, **S. Son** (2024) Understanding Tidal Jet Vortices over Complex Bathymetry via Numerical Modeling and Drone Observation: Match and Mismatch in the Vortex Dynamics under Idealized and Realistic Topographic Settings, ***Journal of Geophysical Research: Oceans***, 129(12), e2024JC021523.

J36. X. Qian†, S. Hwang†, **S. Son** (2024) A Study on Key Determinants in Enhancing Storm Surges along the Coast: Interplay between Tropical Cyclone Motion and Coastal Geometry, ***Journal of Geophysical Research: Oceans***, 129(2), e2023JC020400.

J35. S. Hwang†, **S. Son** (2023) An Efficient HLL-based Scheme for Capturing Contact Discontinuity in Scalar Transport by Shallow Water Flow, ***Communications in Nonlinear Science and Numerical Simulation***, 127, 107531.

J34. **S. Son**, T. Jung (2022) Statistical Analysis on the Tsunamis from Multiple Faults' Sequential Failure with Different Time Intervals and Geographical Layouts, ***Ocean Engineering***, 250, 110720.

J33. Y. Na†, B. Na†, **S. Son** (2022) Near Real-time Predictions of Tropical Cyclone Trajectory and Intensity in the Northwestern Pacific Ocean Using Echo State Network, ***Climate Dynamics***, 58(3), 651–667.

J32. I. Yeo, T. Jung, **S. Son**, H. Yoon (2022) Probabilistic Assessment of Delayed Multi-Fault Rupture Effect on Maximum Tsunami Runup along the East Coast of Korea, ***KSCE Journal of Civil Engineering***, 26(1), 1–12.

J31. B. Na†, **S. Son** (2021) Void Fraction Estimation Using a Simple Combined Wave Gauge System under Breaking Waves, ***Ocean Engineering***, 241, 110059.

J30. S. Tavakkol, **S. Son**, P. Lynett (2021) Adaptive Third Order Adams–Bashforth Time Integration for Extended Boussinesq Equations, ***Computer Physics Communications***, 265, 108006.

J29. B. Na†, **S. Son** (2021) Prediction of Atmospheric Motion Vectors around Typhoons Using Generative Adversarial Network, ***Journal of Wind Engineering and Industrial Aerodynamics***, 214, 104643.

J28. B. Na†, **S. Son**, J. C. Choi (2021) Modeling of Accidental Oil Spills at Different Phases of LNG Terminal Construction, ***Journal of Marine Science and Engineering***, 9(4), 392.

J27. Y. Kim, **S. Son**, T. Jung, T. Gallien (2021) An Analytical and Numerical Study of a Vertically Discretized Multi-paddle Wavemaker for Generating Free Surface and Internal Waves, ***Coastal Engineering***, 165, 103840.

J26. T. Jung, **S. Son** (2021) Active Tsunami Generation by Tectonic Seafloor Deformation of Arbitrary Geometry Considering the Rupture Kinematics, ***Wave Motion***, 100, 102683.

J25. S. Hwang†, **S. Son**, C. Lee†, H. Yoon (2020) Quantitative Assessment of Inundation Risks from Physical Contributors Associated with Future Storm Surges: A Case Study of Typhoon Maemi (2003), ***Natural Hazards***, 104(2), 1389–1411.

J24. J.-A. Yang†, S. Kim, **S. Son**, N. Mori, H. Mase (2020) Assessment of Uncertainties in Projecting Future Changes to Extreme Storm Surge Height Depending on Future SST and Greenhouse Gas Concentration Scenarios, ***Climatic Change***, 162(2), 425–442.

J23. J. Borrero, T. Solihuddin, H. Fritz, P. Lynett, G. Prasetya, V. Skanavis, S. Husrin, W. Kongko, D. Istiyanto, A. Daulat, D. Purbani, H. Salim, R. Hidayat, V. Asvaliantina, M. Usman, A. Kodijat, **S. Son**, C. Synolakis (2020) Field Survey and Numerical Modelling of the December 22, 2018 Anak Krakatau Tsunami, ***Pure and Applied Geophysics***, 177(6), 2457–2475.

J22. Y. Seo, H.-S. Ko, **S. Son** (2020) The Effect of Nozzle Geometry on the Turbulence Evolution in an Axisymmetric Jet Flow: A Focus on Fractals, ***Physica A: Statistical Mechanics and its Applications***, 550, 124145.

J21. **S. Son**, P. Lynett, A. Ayca (2020) Modeling Scour and Deposition in Harbors Due to Complex Tsunami-Induced Currents, ***Earth Surface Processes and Landforms***, 45(4), 978–998.

J20. P. Lynett, D. Swigler, H. El-Safty, L. Montoya, A. Keen, **S. Son**, P. Higuera (2019) Three-dimensional Hydrodynamics Associated with a Solitary Wave Traveling over an Alongshore-Variable, Shallow Shelf, ***ASCE Journal of Waterway, Port, Coastal, and Ocean Engineering***, 145(6), 04019024.

J19. M. Cho, H.-D. Yoon, K. Do, **S. Son**, I.-H. Kim (2019) Comparative Study on the Numerical Simulation of Bathymetric Changes under Storm Conditions, ***Journal of Coastal Research***, SI91, 106–110.

J18. D.-H. Kim, **S. Son** (2019) Role of Shelf Geometry and Wave Breaking in Single N-type Tsunami Runup under Geophysical Scale, ***Ocean Modelling***, 138, 13–22.

J17. C. Lee†, S. Hwang†, K. Do, **S. Son** (2019) Increasing Flood Risk due to River Runoff in the Estuarine Area during a Storm Landfall, ***Estuarine, Coastal and Shelf Science***, 221, 104–118.

J16. D.-H. Kim, **S. Son** (2018) Lagrangian-like Volume Tracking Paradigm for Mass, Momentum and Energy of Nearshore Tsunami and Damping Mechanism, ***Scientific Reports***, 8(1), 14183.

J15. Y. Seo, H.-S. Ko, **S. Son** (2018) Multifractal Characteristics of Axisymmetric Jet Turbulence Intensity from RANS Numerical Simulation, ***Fractals***, 26(1), 1850008.

J14. **S. Son**, Y.-U. Ryu, T.-H. Jung (2017) An Energy-controlling Boundary Condition for Partial Wave Reflections in the Mild Slope Equation, ***Applied Ocean Research***, 68, 244–248.

J13. T.-H. Jung, **S. Son**, Y. Ryu (2017) Finite Element Solution of Linear Waves on a Sloping Bottom Boundary, ***Journal of Coastal Research***, 33(3), 731–737.

J12. **S. Son**, J. Kim, H.-D. Yoon, T.-H. Jung, K. Do, S. Shin (2017) An Observational and Numerical Study of Storm-Induced Morphologic Changes at Sanpo Beach, Korea, ***Journal of Coastal Research***, SI79, 334–338.

J11. S. Shin, J. Nam, **S. Son**, I.-H. Kim, T.-H. Jung (2017) Field Observation and Numerical Modelling of Rip Currents within a Pocket Beach, ***Journal of Coastal Research***, SI79, 229–233.

J10. H.-D. Yoon, M. Cho, **S. Son** (2016) Investigation of Wave Breaking Turbulence in Morphodynamic Modelling, ***Journal of Coastal Research***, SI75, 942–946.

J9. **S. Son**, T.-H. Jung, F. Shi (2016) Vertical Structure of Rip Currents in the Nearshore Circulation, ***Journal of Coastal Research***, SI75, 1402–1406.

J8. T.-H. Jung, **S. Son**, P. Lynett (2016) A Comprehensive Sensitivity Analysis of Tsunami Model System to the Parametric and Input Uncertainties, ***Journal of Coastal Research***, SI75, 1117–1121.

J7. **S. Son**, P. Lynett (2014) Interaction of Dispersive Water Waves with Weakly Sheared Currents of Arbitrary Profile, ***Coastal Engineering***, 90, 64–84.

J6. **S. Son**, P. Lynett (2014) Nonlinear and Dispersive Free Surface Waves Propagating over Fluids with Weak Vertical and Horizontal Density Variation, ***Journal of Fluid Mechanics***, 748, 221–240.

J5. P. Lynett, J. Borrero, **S. Son**, R. Wilson, K. Miller (2014) Assessment of the Tsunami-Induced Current Hazard, ***Geophysical Research Letters***, 41(6), 2048–2055.

J4. P. Lynett, R. Weiss, W. Renteria, G. Morales, **S. Son**, M. Arcos, B. MacInnes (2013) Coastal Impacts of the March 11th Tohoku, Japan Tsunami in the Galapagos Islands, ***Pure and Applied Geophysics***, 170(6–8), 1189–1206.

J3. P. Lynett, J. Borrero, R. Weiss, **S. Son**, D. Greer, W. Renteria (2012) Observations and Modeling of Tsunami-Induced Currents in Ports and Harbors, ***Earth and Planetary Science Letters***, 327–328, 68–74.

J2. **S. Son**, P. Lynett, D.-H. Kim (2011) Nested and Multi-Physics Modeling of Tsunami Evolution from Generation to Inundation, ***Ocean Modelling***, 38, 96–113.

J1. S.-H. Oh, K.-D. Suh, **S.-Y. Son**, D.-Y. Lee (2009) Performance Comparison of Spectral Wave Models Based on Different Governing Equations Including Wave Breaking, ***KSCE Journal of Civil Engineering***, 13(2), 75–84.

Manuscripts under Review

R1. J. Kim, M.-J. Koh†, S. Oh, **S. Son** A Factorized Fourier Neural Operator Surrogate for Basin-Scale Tsunami Propagation, ***Geoscientific Model Development*** (in revision).

R2. M.-J. Koh†, **S. Son** Spatial Redistribution of Probabilistic Tsunami Hazard under Sea-Level Rise: Shoaling–Refraction Decomposition along Korea's East Coast, ***Earth's Future*** (in revision).

R3. A. Chrysanti†, **S. Son** Migration and Intensification as Spatially Separable Drivers of Future Storm Surge Hazard along the Western North Pacific Margin, ***Journal of Geophysical Research: Oceans*** (in review).

R4. X. Qian†, **S. Son** A Probabilistic Wind–Rainfall Dependence Framework with Adaptive Conditional Sampling for Parametric Tropical Cyclone Rainfall Modeling, ***Journal of Hydrology*** (in review).

R5. V. Villanueva†, J. Kang†, **S. Son** Coupling Pre-Trained AI Weather Forecasts with Hydrodynamic and Lagrangian Models for Rapid Storm Surge and Marine Debris Prediction, ***Ocean Modelling*** (in review).

R6. S. Im†, S. Hwang, J. Youn†, **S. Son** Reconstructing a Tropical Cyclone Wind Field Using Refined Satellite-Derived Atmospheric Motion Vectors and Its Application to Storm Surge Modeling, ***IEEE Transactions on Geoscience and Remote Sensing*** (in review).

R7. J. Kim, et al., **S. Son** Robustness of Projected Directional Design Wave Heights to Extreme-Value Model Choice and Typhoon Influence along the Korean Coast, ***Ocean Engineering*** (in review).

R8. T.-H. Jung, **S. Son** A Time-Domain Boussinesq-Type Model for Wave Propagation over a Viscous Mud Layer with Variable Water Depth, ***Ocean Engineering*** (in review).

R9. T.-H. Jung, H. Park, **S. Son** Wave Diffraction by a Partially Reflecting Semi-Infinite Breakwater: An Exact Wiener–Hopf Solution, ***Applied Ocean Research*** (in review).

R10. T.-H. Jung, **S. Son** Discrete-Adjoint Reconstruction of the Initial Tsunami Free-Surface Field Using an Observability-Based Gauge Network, ***Ocean Modelling*** (submitted).

Conference Proceedings and Presentations (* denotes oral presenter)

C1. **S. Son***, S. Im (2026) Improving Storm Surge Predictions Using Satellite-derived Atmosphere Motion Vectors, *39th International Conference on Coastal Engineering*, Galveston, TX, USA.

C2. J. Kim*, **S. Son** (2026) Boundary-informed Deep Learning SWAN Emulator for Nearshore Wave Forecasting, *39th International Conference on Coastal Engineering*, Galveston, TX, USA.

C3. J. Kang*, **S. Son** (2026) GPU-based Tsunami MOdeling for Basin-scale Simulation, *39th International Conference on Coastal Engineering*, Galveston, TX, USA.

C4. M. Kim, **S. Son** (2026) Long-term Wave Projections under a HIgh-emission Scenarios via Extreme Oriented correction (poster), *39th International Conference on Coastal Engineering*, Galveston, TX, USA.

C5. J. Kim*, **S. Son** (2026) Emulating SWAN for Operational Coastal Waves with UNet++ and ConvLSTM, *IAHR Asia-Pacific Division Congress*, Songdo, Incheon, Korea.

C6. M. Kim*, **S. Son** (2026) Improving Long-Term Wave Projections via Extreme-Oriented Bias Correction, *IAHR Asia-Pacific Division Congress*, Songdo, Incheon, Korea.

C7. S. Oh*, M. Koh, **S. Son** (2026) Logic-Tree Based Probabilistic Tsunami Hazard Assessment and Deaggregation for the Eastern Coast of Korea, *IAHR Asia-Pacific Division Congress*, Songdo, Incheon, Korea.

C8. **S. Son***, S. Im (2026) Improving Tropical Cyclone Wind and Pressure Field Reconstruction Using GK-2A Atmospheric Motion Vectors and Machine Learning, *EGU General Assembly 2026*, Vienna, Austria.

C9. S. Oh, M.-J. Koh, **S. Son** (2026) Probabilistic Tsunami Hazard Assessment and Deaggregation for the Eastern Coast of Korea (poster), *EGU General Assembly 2026*, Vienna, Austria.

C10. J. Kang*, **S. Son** (2026) A GPU-accelerated Framework for Basin-Scale Tsunami Propagation in Early Warning Applications, *EGU General Assembly 2026*, Vienna, Austria.

C11. X. Qian*, **S. Son** (2025) Impact of Typhoon Rainfall from Different Landfall Paths on Coastal Compound Flooding, *International Conference on Asian and Pacific Coasts (APAC 2025)*, Busan, Korea.

C12. **S. Son***, X. Qian (2025) Assessing the Impact of Tropical Cyclone Rainfall Motion on Coastal Compound Flooding, *4th International Workshop on Waves, Storm Surges and Coastal Hazards (18th International Waves Workshop)*, Santander, Spain.

C13. X. Qian, **S. Son** (2024) Examining the Role of Tropical Cyclone Rainfall Motion in Coastal Compound Flooding, *38th International Conference on Coastal Engineering*, Rome, Italy.

C14. J. Youn*, **S. Son** (2024) The Spatial Distribution of the Tropical Cyclone Pressure Using Machine Learning Technique, *38th International Conference on Coastal Engineering*, Rome, Italy.

C15. S. Hwang, **S. Son** (2024) A GPU-accelerated Numerical Model for Coastal Flooding, *38th International Conference on Coastal Engineering*, Rome, Italy.

C16. J. Noh*, **S. Son** (2024) Morphological Change Simulation Using a GPU-based Platform, *38th International Conference on Coastal Engineering*, Rome, Italy.

C17. **S. Son**, X. Qian, S. Hwang (2023) Understanding the Role of Cyclone Kinematics and Coastal Geometry in Developing Surge Levels along the Coast, *AGU Fall Meeting 2023*, San Francisco, CA, USA.

C18. S. Im*, **S. Son** (2023) Improving the Predictability of Storm Surge Modelling Using Satellite Image-based Wind–Pressure Distribution, *AGU Fall Meeting 2023*, San Francisco, CA, USA.

C19. J. Noh, **S. Son** (2023) GPU-based Real-time Interactive Sediment Transport Model (poster), *AGU Fall Meeting 2023*, San Francisco, CA, USA.

C20. S. Hwang, **S. Son** (2023) Immersive Hydrodynamic Model for Flood Simulation with Virtual Reality Simulation Environment (poster), *AGU Fall Meeting 2023*, San Francisco, CA, USA.

C21. **S. Son***, X. Qian (2023) On the Effect of Tropical Cyclones' Translation Speed and Landfall Trajectory on the Storm Surge Dynamics, *EGU General Assembly 2023*, Vienna, Austria.

C22. X. Qian*, **S. Son** (2022) The Effect of Tropical Cyclones' Translation Speeds and Landfall Angles on Maximum Surge Heights along Idealized Coasts, *37th International Conference on Coastal Engineering*, Sydney, Australia.

C23. S. Hwang*, P. Lynett, **S. Son** (2022) Nearshore Scalar Transport Model with Virtual Reality Environment, *37th International Conference on Coastal Engineering*, Sydney, Australia.

C24. A. Chrysanti, **S. Son** (2022) Hydro-meteorological Analysis of Extreme Weather in Indonesia Using Coupled Atmospheric–Hydrological Model (poster), *37th International Conference on Coastal Engineering*, Sydney, Australia.

C25. S. Hwang*, **S. Son** (2021) Development of GPU-accelerated 2DH Hydrodynamic and Transport Model, *9th International Symposium on Environmental Hydraulics*, Seoul, Korea.

C26. **S. Son***, P. Lynett (2020) Inter-Coupled Tsunami Modelling through an Absorbing-Generating Boundary, *Virtual International Conference on Coastal Engineering (vICCE 2020)*, Online.

C27. S. Hwang, **S. Son**, P. Lynett (2020) A GPU-accelerated Modeling of Scalar Transport Based on Boussinesq-type Equations, *Virtual International Conference on Coastal Engineering (vICCE 2020)*, Online.

C28. J.-A. Yang, S. Kim, **S. Son**, H. Mase, N. Mori (2020) Uncertainty Assessment of Future Change Projection to Typhoon Properties and Maximum Storm Surge Height Depending on Future Sea Surface Temperature and Greenhouse Gas Emission Scenarios, *Virtual International Conference on Coastal Engineering (vICCE 2020)*, Online.

C29. **S. Son**, T. Jung, D.-H. Kim, H.-D. Yoon (2020) Modeling Morphological Changes by Tsunami-Induced Currents, *EGU General Assembly 2020*, Vienna, Austria.

C30. J. Yang, **S. Son** (2019) Parameter Sensitivity Analysis of Delft3D in the Assessment of Hydrodynamics and Sediment Transport at Maengbang Beach, *Coastal Structures 2019*, Hannover, Germany.

C31. M. Cho, J. Yang, **S. Son**, H. Yoon (2019) Sensitivity and Uncertainty Analysis of Coastal Numerical Model under Various Beach Conditions in Korea, *Coastal Sediments 2019*, St. Petersburg, FL, USA.

C32. Y. Na, **S. Son** (2018) Prediction of Tropical Cyclone Trajectories Using Echo State Networks, *AGU Fall Meeting*, Washington, DC, USA.

C33. **S. Son***, C. Lee, K. Do, T. Jung (2018) Coupled Hydrodynamic–Hydrological Modeling for Storm Surge Inundation in the Coastal Area, *36th International Conference on Coastal Engineering*, Baltimore, MD, USA.

C34. **S. Son**, C. Lee, K. Do, T. Jung (2018) Establishment of Coupled Modelling System by Linking Storm-Surge, Tide and River Physics, *Ocean Sciences Meeting 2018*, Portland, OR, USA.

C35. S. Hwang, **S. Son** (2018) 3D Numerical Investigation on the Onset and Growth of Tide-induced, Geometry-governed Turbulent Coherent Structure: A Case Study of Uldolmok Strait, Korea, *3rd International Water Safety Symposium*, Incheon, Korea.

C36. **S. Son***, C. Lee (2018) Inland Flooding Responses to the Inclusion of Estuarine Discharges in the Storm Surge Modelling, *28th International Ocean and Polar Engineering Conference*, Sapporo, Japan.

C37. D.-H. Kim*, **S. Son** (2018) Mass, Momentum and Energy Paradigm of Idealized Tsunami: On Steep Sloped Bathymetry, *28th International Ocean and Polar Engineering Conference*, Sapporo, Japan.

C38. **S. Son**, C. Lee (2017) Establishment of Coupled Modelling System by Linking Surge–Tide–Riverine Flow Effect, *International Research Symposium on Engineering and Technology*, Singapore.

C39. **S. Son***, C. Lee, K. Do, T. Jung (2017) Increasing Flood Risk due to Run-off Outflow near Estuarine City during Storm Event, *AGU Fall Meeting*, New Orleans, LA, USA.

C40. **S. Son***, P. Lynett (2017) Tsunami Modelling System Inter-coupled through Absorbing-Generating Boundaries, *27th International Tsunami Symposium*, Bali, Indonesia.

C41. T. Jung, **S. Son** (2016) Effective Boundary Condition for Wave Treatment, *12th International Conference on Hydroinformatics*, Incheon, Korea.

C42. **S. Son**, T. Jung (2016) Tsunami Modelling System Coupled through Absorbing-Generating Boundaries, *12th International Conference on Hydroinformatics*, Incheon, Korea.

C43. **S. Son** (2015) Coupling Efforts to the Accurate and Efficient Tsunami Modelling System, *AGU Fall Meeting 2015*, San Francisco, CA, USA.

C44. **S. Son**, T.-H. Jung (2015) 2DH Analysis of Morphologic Responses to the Tsunami Impacts Using Process-based Model, *8th International Conference on Asian and Pacific Coasts (APAC 2015)*, Chennai, India (*Procedia Engineering*, 116, 979–985).

C45. D.-H. Kim*, **S. Son** (2015) Coastal Geomorphology Model Using Boussinesq Equation Considering Undertow Effect, *25th International Ocean and Polar Engineering Conference*, Kona, HI, USA.

C46. **S. Son***, P. Lynett (2014) Numerical Prediction of Tsunami-Induced Sediment Transport in the Harbor, *AGU Fall Meeting*, San Francisco, CA, USA.

C47. **S. Son***, P. Lynett (2014) Numerical Study of Morphological Changes by Far-Field Tsunami Impacts, *34th International Conference on Coastal Engineering*, Seoul, Korea.

C48. P. Lynett*, J. Borrero, **S. Son**, R. Wilson, K. Miller (2013) Assessment of Nearshore Hazard due to Tsunami-Induced Currents (Invited), *AGU Fall Meeting*, San Francisco, CA, USA.

C49. P. Lynett*, J. Borrero, **S. Son**, R. Wilson, K. Miller (2013) Assessment of Near Shore Hazard due to Tsunami-Induced Currents, *26th International Tsunami Symposium*, Göcek, Turkey.

C50. T.-H. Jung, M. Son, **S. Son**, H.-S. Park (2013) Inclined Bottom Boundary Condition for the Mild-slope Equation, *7th International Conference on Asian and Pacific Coasts*, Bali, Indonesia.

C51. P. Lynett*, J. Borrero, R. Wilson, K. Miller, **S. Son** (2013) Detailed Simulation of Tsunami-Induced Currents in California Ports and Harbors, *Ports 2013*, Seattle, WA, USA.

C52. P. Lynett*, **S. Son**, D.-H. Kim (2011) Simulation of Tsunami-Induced Currents in Ports and Harbors, *Coastal Structures 2011*, Yokohama, Japan.

C53. P. Lynett*, D. Swigler, **S. Son**, D. Bryant, S. A. Socolofsky (2010) Experimental Study of Solitary Wave Evolution over a 3D Shallow Shelf, *32nd International Conference on Coastal Engineering*, Shanghai, China.

C54. K.-D. Suh*, **S. Son**, J.-I. Lee, T.-H. Lee (2002) Calculation of Irregular Wave Reflection from Perforated-wall Caisson Breakwaters Using a Regular Wave Model, *28th International Conference on Coastal Engineering*, Cardiff, UK.